INCH-POUND

MIL-PRF-27626H (USAF)
27 May 2021
SUPERSEDING
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23 December 2016

PERFORMANCE SPECIFICATION

SAMPLER, CRYOGENIC LIQUID

Inactive for new design after 24 April 1981

This specification is approved for use by the USAF and is available for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cryogenic liquid sampler consisting of a sampling assembly, shipping vessel, and accessories.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Area B, Wright-Patterson AFB, OH 45433-7017 or email to engineering.standards@us.af.mil. Since contact information can change, you may want verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6695

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59860 - Valves, Cylinder, Gas (for Compressed or Liquefied Gases)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27210 - Oxygen, Aviator's Breathing, Liquid and Gas

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of these documents are available from https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 178.69 - Responsibilities and requirements for manufacturers of UN pressure receptacles.

49 CFR 178.70 - Approval of UN pressure receptacles.

49 CFR 178.71 - Specifications for UN pressure receptacles.

(Copies of the CFRs may be obtained at https://www.ecfr.gov/.)

DEPARTMENT OF THE AIR FORCE

7545352 - Requirements for Finishes, Materials, and Processes for Corrosion Prevention and Control on Warner Robins ALC Ground Support Equipment.

(Copies of these drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9809-1 - Gas cylinders -- Refillable seamless steel gas cylinders
ISO 13769 - Gas cylinders -- Stamp marking

(Copies of these documents are available from ISO online at https://www.iso.org.)

SOCIETY OF AUTOMOTIVE ENGINEERING (SAE)

SAE ARP1176 - Oxygen System and Component Cleaning and Packaging

(Copies of these documents are available from SAE online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.4.1 Pressure receptacle acceptability flow chart. There are multiple laws governing the specifications for pressure receptacles and the appropriate markings for each standard. For the sake of clarity, a flow chart has been added to this specification. To determine what receptacle is acceptable anywhere in the world, please refer to Appendix 1.

3. REQUIREMENTS

3.1 Components. The sampler shall consist of the following major components:

Liquid Sampler Assembly
UN Pressure Receptacle (ISO 9809-1)
Sample Extraction Valve Assembly
Valve, Cylinder, Gas (A-A-59860)
Sample Cup Assembly
Hose(s) Assembly
Pressure Relief Device (PRD) / Rupture Disk (A-A-59860)
Pressure Gauge Assembly
Certified ATA 300 Category I, DOT31FP, Air Shipment Case

3.2 Materials. All metallic and nonmetallic materials used in the sampler shall be safe for use with oxygen. All metals used shall be of the corrosion-resistant type or treated to resist corrosion.

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3.2.1 Recycled, virgin, and reclaimed materials. Provided that all other requirements of this specification are met, reclaimed materials shall be used to the maximum extent possible with no exclusion to the use of recovered materials and no requirement that an item be manufactured from virgin materials.

3.3 Design. The sampler shall include all parts and accessories for:

a. Being connected to a cryogenic vessel and obtaining a representative sample of liquid.

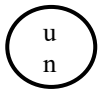
b. Converting the liquid sample to gas, transferring to a pressure vessel, and retaining it without loss of pressure for shipment to an analysis laboratory.

3.3.1 Configuration. The liquid-sample container shall be capable of being precooled to the atmospheric pressure boiling temperature of the liquid being sampled prior to the actual sampling operation. The pressure vessel shall be certified to ISO 9809-1, in accordance with Title 49 of the Code of Federal Regulations (49 CFR).

3.4 International approvals. Per 49 CFR, future acquisitions of receptacles ear-marked for international use shall be manufactured in compliance with two different international standards.

3.4.1 ISO 9809-1 approval. The sampler shall be constructed and certified in accordance with ISO 9809-1. As such, the sampler shall be marked in accordance with ISO 13769. Hereafter, this sampler will be referred to as a UN pressure receptacle. The nation of approval shall be listed as USA, in accordance with 49 CFR 178.71.

3.4.1.1 Figure of ISO 13769. The standard for markings on a UN pressure receptacle is delineated in ISO 13769. Provided the third row of markings looks like figure 1 below, this UN pressure receptacle will be viable for use anywhere in the world, including the USA. See Appendix 1 for a larger diagram of the stamp.



ISO 9809-1 USA/Mxxxx*

* The numbers after the USA/ are a manufacturing mark. This will be an alphanumeric code that will match the code on the approval list. This alphanumeric code will be different for each receptacle, dependent upon which manufacturer produced it and which inspector inspected it. The “Mxxxx” is intended solely as an abbreviation, and should be interpreted as an acceptable manufacturing mark.

FIGURE 1: Third Row Markings for UN Pressure Receptacle

3.5 Performance.

3.5.1 Sample volume. The sampler shall be capable of providing a gaseous sample with a minimum volume of 100 standard liters (at 70 ºF and 760 mm Hg).

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3.5.2 Leak test pressure. The sampler shall be capable of being pressurized to 1600 psig with gaseous oxygen without exhibiting leakage.

3.6 Weight. The weight of the sampler and its accessories shall not exceed 50 pounds. The additional weight is to account for added items from the previous version of this specification.

3.7 Finishes and protective coatings.

3.7.1 Surfaces contacting oxygen. All surfaces, parts, fittings, etc., of the sampler that will be in contact with high-purity oxygen shall be thoroughly cleaned in accordance with procedures found in SAE ARP1176. No other cleaning, priming, or painting with organic materials shall be performed on these surfaces.

3.7.2 Exposed parts and surfaces. All exposed metal parts and surfaces, except parts and surfaces that contact high-purity oxygen, shall be cleaned, treated, and finished per Air Force Drawing 7545352, with the exception that no paint or primer is to be applied to components other than the carrying case.

3.8 Operational markings.

3.8.1 Inlet and outlet fittings. The inlet and outlet fittings shall be clearly identified by markings, either etched or stenciled, directly on the sample bottle.

3.8.2 Instructions. Operating instructions shall be provided on a plate of sheet aluminum or zinc of not less than 0.050-inch thickness, anodized or etched to produce raised markings with a black or other dark color background, and with a border of not less than ¼-inch. The plate shall be attached to the top half of the case by rivets. The plate shall also be located on the side of the sampler case so that the instructions can be read when the case is in an upright position.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. All parts of the sampler shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges.

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.

c. Thoroughness of soldering, welding, brazing, painting, and riveting.

d. Thorough removal of rust, slag, scale, flux, and other foreign materials from inside of the sampler and all other surfaces that contact the oxygen in either liquid or gaseous form.

e. Alignment of parts and tightness of assembly screws, bolts, rivets, etc.

f. For rivet tightness and properly headed.

3.11 General cleaning instructions. Following completion of fabrication and assembly

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operations, the sampler shall be thoroughly cleaned and degreased to remove all contaminating materials. Particular emphasis shall be placed upon complete removal of all traces of cleaning fluids used. The adequacy of the cleaning operation shall be verified by pressurizing the sampler to 1600 psig with gaseous oxygen, which has been analyzed for trace constituents by use of an infrared spectrophotometer. The gas from the cleaned sampler shall be introduced into an infrared spectrophotometer and analyzed to determine if trace constituents were introduced from the sampler. The sampler shall be considered clean when no increase in contaminants is detected.

3.11.1 Cleaning. The sampler shall be thoroughly cleaned to remove excess and spilled lubrication materials, loose or chipped paint, spilled chemicals, and other foreign materials. All cleaning solvents shall be removed from the sampler components prior to delivery.

3.11.1.1 Degreasing. Sampler surfaces, parts, fittings, etc., shall be degreased in accordance with procedures in SAE ARP1176.

3.11.1.1.1 Flammable solvents. Petroleum, petroleum-based, and other flammable solvents shall not be used on such surfaces.

3.11.2 Final cleaning and pressurizing. Following assembly and completion of all testing, the sampler shall be emptied and placed in an oven with a minimum temperature of 250 ºF. While in the oven, the sampler shall be purged with 100 cfm of aviator's breathing oxygen or dry, oil-free nitrogen at 250 ºF. Upon completion of the purging operations, the unit shall be evacuated to an absolute pressure of 100 microns Hg while still in the oven. The sampler shall then be removed from the oven, the vacuum broken, and the sampler pressurized with dry, oil-free, gaseous aviator's breathing oxygen to a pressure of 25 psig and all openings be closed. Tag sampler stating that it is clean and pressurized with clean, dry, oil-free, aviator's breathing oxygen. Tag shall be securely attached to the sampler valve handle.

4. VERIFICATION

4.1 Classification of tests. The inspection and testing of the sampler shall be classified as follows:

a. Individual tests (see 4.3).

4.2 Test conditions.

4.2.1 Apparatus. To the extent as practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision-type, and shall be calibrated at intervals properly spaced to ensure continuous laboratory accuracy.

4.2.2 Pressure-gage accuracy. Data on gage pressure shall be accurate to within 10 percent.

4.3 Individual tests. Each sampler shall be subjected to the following test as described under 4.5:

a. Examination of product (see 4.4.1).

b. Mechanical inspection (see 4.4.2).

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c. Functional check (see 4.4.3).

d. Pressure test (see 4.4.4).

e. Individual operational test (see 4.4.5).

4.4 Test methods.

4.4.1 Examination of product. The sampler shall be examined to determine compliance with this specification with respect to materials, workmanship, marking, and as specified herein.

4.4.2 Mechanical inspection. A mechanical inspection of all components and parts shall be conducted. All pertinent data concerning conditions, defects of manufacture, damage in transit, and damage through use prior to test shall be recorded.

4.4.3 Functional check. All mechanical parts shall be checked for free and proper functioning.

4.4.4 Pressure test. The sampler shall be pressure-tested and certified in accordance with 49 CFR 178.69 and 178.70.

4.4.5 Individual operational test. Each sampler shall be connected to a source of liquid oxygen. The sampler shall be used in accordance with the operational instructions to obtain a liquid sample. The sampler shall then be disconnected from the liquid source and the sample allowed to vaporize. A resulting vessel pressure indication of less than 800 psig within 20 minutes from the time the sample was taken shall be cause for rejection. The vessel shall then remain pressurized for at least 24 hours. Any loss of pressure from the initial fill pressure that is attributable to leakage shall be cause for rejection.

4.4.6 Cleaning effectiveness test. One unit from each lot of 25, or fraction thereof, shall be selected at random from the completed and cleaned samplers that are ready for shipment. The sampler shall be pressurized to 1600 psig with gaseous oxygen or nitrogen which has been analyzed for trace constituents by use of an instrument or instruments capable of determining trace constituents at the level specified in MIL-PRF-27210. The gas from the sampler shall then be introduced into the same analytical equipment to determine if any trace constituents were introduced from the sampler. If any additional trace constituents or an increase in the level of any of the original contaminants is detected, the sampler shall be rejected and re-cleaned. Any rejected samplers shall be individually checked after re-cleaning and will be accepted only upon proving they are clean. In the event that a sampler from a given lot fails to meet the cleanliness requirements, three additional samplers shall be selected at random from the same lot and tested for cleanliness in the same manner. If any of the three additional samplers is rejected, the complete lot shall be rejected, re-cleaned, and subjected to the test again. Any lots which have been rejected shall have five samplers selected at random and subjected to the cleanliness test. If any of the five additional samplers selected fails to meet the requirements of 3.11, each sampler in the lot shall be subjected to the test and accepted only upon satisfactory completion of the test.

4.5 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of section 5 or the documents specified therein.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's System Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sampler is intended for obtaining representative samples of liquid oxygen or liquid nitrogen to permit close quality control of the liquid oxygen or nitrogen. This is a military-unique requirement.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Level of protection.

c. Requested quantity of UN pressure receptacle assemblies.

d. Requested quantity of ATA 300 air shipment cases.

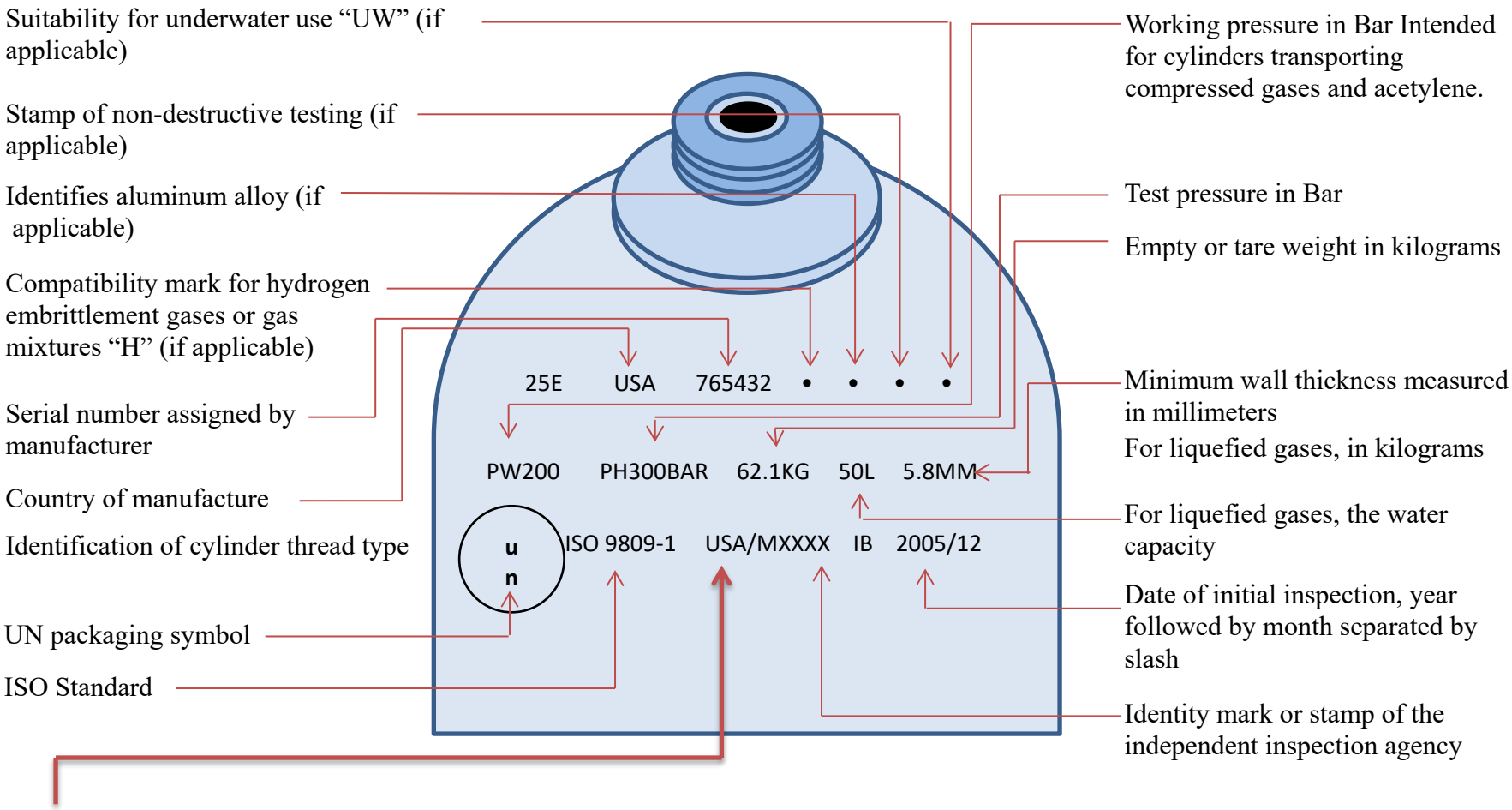
6.3 Subject term (key word) listing.

Aviator's breathing oxygen
Liquid nitrogen
Liquid oxygen
Sampling assembly

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Current Accepted Marking Guidance



Mark in question: IAW with ISO 9809-1 (and subsequently USDOT 49 CFR Part 178.71 and ADR, Vol II, Chapter 6.2.2, the only acceptable marking is the abbreviation for the nation in which the bottle was inspected and accepted as falling in line with the specifications outlined in ISO 9809-1 by the on-site inspector.

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CONCLUDING MATERIAL

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Preparing activity:
Air Force – 184

Review activities:
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Agent:
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